

EEEM071 - ADVANCED TOPICS IN COMPUTERVISION AND DEEP LEARNING VEHICLE RE-IDENTIFICATION - COURSEWORK REPORT

Module	Advanced Topics in Computer Vision and Deep Learning – EEEM071
Assessment Type	Coursework Assignment
Weighting	25%
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INTRODUCTION

Vehicle Re-Identification (Re-ID) is the computer vision task of matching vehicle images captured across separate, non-overlapping cameras. Unlike standard object detection or classification, Re-ID demands fine-grained discriminative representations that remain robust under dramatic variations in viewpoint, illumination, scale, and occlusion. A vehicle photographed side-on under bright daylight at one camera must be correctly matched to its darker, front-facing image at another — even when hundreds of visually similar vehicles populate the gallery.

The practical importance of vehicle Re-ID is considerable. Modern intelligent transport systems deploy surveillance cameras across city-wide networks, yet without Re-ID each camera operates in isolation. Effective Re-ID enables authorities to track individual vehicles across camera boundaries, reconstructing travel routes, monitoring congestion, identifying vehicles of interest, and enforcing traffic regulations — capabilities that are central to law enforcement, border security, and smart city infrastructure.

This coursework presents a systematic CNN-based investigation on the VeRi benchmark dataset across three experimental dimensions: (1) the influence of backbone architecture on retrieval performance, (2) the impact of data augmentation strategies, and (3) model sensitivity to key training hyperparameters — learning rate, batch size, and optimiser. All experiments were conducted on Google Colab using an NVIDIA Tesla T4 GPU. The best result achieved was 59.3% mAP and 86.2% Rank-1 with ResNet-50 at LR=0.0001, representing a 13.7 percentage point mAP gain over the MobileNetV3-Small baseline.

SECTION 1: FAMILIARITY WITH THE CODE PROVIDED

Question 1.1 – Default Settings (MobileNetV3-Small)

The pipeline starts by loading the VeRi dataset, which contains 37,778 training images of 576 vehicle identities captured across 20 cameras. During training, images are resized to 224×224 and passed through a MobileNetV3-Small backbone initialised with pretrained ImageNet weights. The training loop runs for 10 epochs using the AMSGrad optimiser at a learning rate of 0.0003.

Two loss functions run jointly throughout training. Cross-entropy loss (Xent) supervises identity classification by pushing each image towards its correct vehicle class. Triplet loss with hard mining (Htri) enforces metric learning by ensuring that matched-vehicle pairs are closer in feature space than mismatched ones. Both losses are combined with equal weighting ($\lambda = 1$ each), giving the network both discriminative class boundaries and a well-structured embedding space.

At evaluation, features extracted from query and gallery sets are compared using cosine distance and ranked. The model achieved mAP of 45.6% and Rank-1 accuracy of 81.5%. These results suggest the lightweight MobileNetV3-Small backbone captures vehicle appearance effectively for a first baseline, though the moderate mAP indicates room for improvement, likely due to the architecture's limited capacity compared to deeper networks. The consistent reduction in Xent across epochs confirms stable convergence.

Architecture	mAP	Rank-1	Rank-5	Rank-10
MobileNetV3-Small (default)	45.6%	81.5%	91.3%	94.6%

Question 1.2 – ResNet-50 (Different CNN Variant)

For this experiment, the backbone was swapped from MobileNetV3-Small to ResNet-50, a well-established deeper architecture. ResNet-50 uses residual skip connections across 50 layers and outputs a 2048-dimensional feature vector, compared to MobileNetV3-Small's 576-dimensional output. All other settings remained unchanged.

The change had a noticeable positive effect. mAP improved from 45.6% to 52.4%, and Rank-1 climbed from 81.5% to 83.8%. This improvement is expected given that ResNet-50's greater depth and wider feature representations capture more discriminative cues for vehicle matching — particularly across viewpoint and lighting changes. The residual connections also help gradients flow during training, leading to better feature convergence.

Training time increased substantially, taking approximately 1 hour and 20 minutes versus 27 minutes for MobileNetV3-Small. This highlights the accuracy-efficiency trade-off between the two

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architectures. For deployment in resource-constrained environments, MobileNetV3-Small may still be preferred, but for maximum retrieval accuracy ResNet-50 is clearly superior at this training scale.

Architecture	mAP	Rank-1	Rank-5	Rank-10
MobileNetV3-Small	45.6%	81.5%	91.3%	94.6%
ResNet-50	52.4%	83.8%	92.4%	95.2%

Question 1.3 – MobileNetV3-Large (Different Network Family)

MobileNetV3-Large was selected as the third architecture. While MobileNetV3-Small and ResNet-50 were used above, MobileNetV3-Large represents a distinct operating point within the MobileNet family — significantly wider and more expressive than its small counterpart, yet substantially more efficient than ResNet-50. This makes for a meaningful three-way comparison.

MobileNetV3-Large achieved mAP of 49.1% and Rank-1 of 83.3%. This places it between MobileNetV3-Small and ResNet-50 in terms of mAP, yet it nearly matches ResNet-50 on Rank-1. The 960-dimensional feature output indicates a richer representation than MobileNetV3-Small's 576 dimensions, which explains the improvement. Training completed in approximately 31 minutes, making it considerably faster than ResNet-50.

Comparing all three architectures, ResNet-50 leads on mAP (52.4%) thanks to its greater depth and residual structure. However, MobileNetV3-Large offers a strong balance — a Rank-1 of 83.3% closely approaching ResNet-50 (83.8%) at roughly one-third of the training time. This suggests MobileNetV3-Large is a practical choice when computational resources are limited but competitive performance is still required.

Architecture	mAP	Rank-1	Rank-5	Rank-10
MobileNetV3-Small	45.6%	81.5%	91.3%	94.6%
MobileNetV3-Large	49.1%	83.3%	92.1%	95.3%
<u>ResNet-50 (best)</u>	<u>52.4%</u>	<u>83.8%</u>	<u>92.4%</u>	<u>95.2%</u>

SECTION 2: DATASET PREPARATION AND AUGMENTATION EXPERIMENT

Question 2.1 – Additional Augmentation Techniques

Building on the default pipeline (random horizontal flip and Random2DTranslation), two additional augmentation techniques were tested one at a time using ResNet-50 as the backbone, which was the best-performing model from Section 1.

Experiment A added Colour Jitter (random brightness, contrast, saturation, and hue perturbations). The idea is that vehicles appear under varied lighting conditions and camera sensors, so training with colour-varied versions should improve robustness. Results: mAP 51.8%, Rank-1 83.8%. Compared to the default ResNet-50 baseline (mAP 52.4%, Rank-1 83.8%), the mAP dropped very slightly. This suggests Colour Jitter introduces some training noise at only 10 epochs, where the model may not yet have had enough time to fully benefit from colour invariance.

Experiment B added Random Erasing (randomly masking rectangular patches from the image). This simulates occlusion, a common challenge in vehicle Re-ID where other vehicles or infrastructure may partially obscure the target. Results: mAP 50.8%, Rank-1 81.8%. Random Erasing reduced both metrics compared to the baseline, suggesting that at 10 epochs its regularisation effect acts more as a harmful distractor than a benefit, as the model struggles to learn discriminative features from heavily occluded training images.

Augmentation Setting	mAP	Rank-1	Rank-5	Rank-10
<u>Default (Flip + Random2DTranslation)</u>	<u>52.4%</u>	<u>83.8%</u>	<u>92.4%</u>	<u>95.2%</u>
Default + Colour Jitter	51.8%	83.8%	92.8%	95.5%
Default + Random Erasing	50.8%	81.8%	92.0%	95.5%

Question 2.2 – Best-Performing Augmentation Combination

The final experiment combined both Colour Jitter and Random Erasing on top of the default augmentation pipeline. Combining techniques can sometimes yield additive benefits if they target different sources of variation independently.

Results: mAP 52.5%, Rank-1 83.6%. The combined setting produced the highest mAP (52.5%) of any augmentation configuration, slightly surpassing the baseline (52.4%) and the individual additions.

This suggests that even though each technique alone reduced performance slightly, using both together achieves marginal complementary regularisation that nudges mAP upward. Rank-1 at 83.6% is marginally below the baseline (83.8%), indicating a minor trade-off.

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Overall, augmentation effects at 10 epochs are modest. With longer training, the benefits of colour and occlusion invariance would likely be more pronounced. The combined configuration is selected as the best augmentation setting for subsequent experiments.

Augmentation Setting	mAP	Rank-1	Rank-5	Rank-10
Default (Flip + Random2DTranslation)	52.4%	83.8%	92.4%	95.2%
Default + Colour Jitter	51.8%	83.8%	92.8%	95.5%
Default + Random Erasing	50.8%	81.8%	92.0%	95.5%
<u>Combined (Default + CJ + RE) [BEST]</u>	<u>52.5%</u>	<u>83.6%</u>	<u>92.4%</u>	<u>95.5%</u>

SECTION 3: EXPLORATION OF HYPERPARAMETERS

Question 3.1 – Learning Rate Exploration

Five learning rate values were tested using ResNet-50 with the default augmentation setting. The default LR is 0.0003. Four additional values — 0.000001, 0.00001, 0.0001, and 0.003 — were chosen to span several orders of magnitude, probing underfitting at low rates and instability at high rates.

LR = 0.000001 produced very poor results (mAP 15.9%, Rank-1 43.8%), as the model barely learned anything in 10 epochs — Xent remained close to 5.9 with training accuracy of only 3%. LR = 0.00001 showed partial convergence (mAP 42.0%, Rank-1 69.6%), but clearly underfitted. LR = 0.0001 delivered the strongest performance: mAP 59.3%, Rank-1 86.2%, a large improvement over the default. Training accuracy reached 96.18% with Xent dropping to 0.20, confirming good convergence. LR = 0.003 caused divergence (mAP 22.6%, Rank-1 46.7%), with high loss oscillation throughout.

The relationship shows a clear inverted-U pattern. LR = 0.0001 is optimal here, striking the right balance between fast learning and stable convergence within 10 epochs. It outperforms the original default LR (0.0003) by over 13 percentage points in mAP, making it the selected value going forward.

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Learning Rate	mAP	Rank-1	Rank-5	Rank-10
0.000001	15.9%	43.8%	62.2%	70.1%
0.00001	42.0%	69.6%	85.1%	90.3%
0.0003 (default)	52.4%	83.8%	92.4%	95.2%
<u>0.0001 (BEST)</u>	59.3%	86.2%	94.5%	96.5%
0.003	22.6%	46.7%	66.6%	75.5%

Question 3.2 – Batch Size Exploration

Using the best learning rate (0.0001) from Question 3.1, five batch sizes were tested: 16, 32, 64 (default), 96, and 128. Batch size affects gradient estimation quality and the number of update steps per epoch, which in turn influences convergence speed and generalisation.

BS=16 achieved mAP 59.5%, Rank-1 89.2% — competitive mAP but notably high Rank-1, reflecting the benefit of more frequent gradient updates. BS=32 produced the best mAP overall at 60.4% with Rank-1 88.9%, indicating that smaller batches with more update steps per epoch improve the model's metric learning. BS=64 (default) gave mAP 59.3%, Rank-1 86.2%. BS=96 delivered mAP 59.6%, Rank-1 87.1%, and BS=128 yielded mAP 59.5%, Rank-1 86.6%.

Larger batch sizes generally yielded slightly lower performance, which is consistent with the known generalisation gap associated with large-batch training. Smaller batches provide noisier but more diverse gradient updates that can help the model escape sharp minima. BS=32 is selected as the optimal batch size, offering the highest mAP with strong Rank-1.

Batch Size	mAP	Rank-1	Rank-5	Rank-10
16	59.5%	89.2%	95.2%	97.7%
<u>32 (BEST)</u>	<u>60.4%</u>	<u>88.9%</u>	<u>95.3%</u>	<u>97.3%</u>
64 (default)	59.3%	86.2%	94.5%	96.5%
96	59.6%	87.1%	94.5%	97.0%
128	59.5%	86.6%	93.7%	96.7%

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Question 3.3 – Optimizer Exploration (SGD)

Using the best LR (0.0001) and best BS (32) from the previous experiments, the optimiser was changed from AMSGrad to SGD. PyTorch's built-in SGD class was used with default momentum and weight decay settings consistent with the original configuration.

SGD produced mAP 28.8% and Rank-1 55.5%, a significant drop compared to AMSGrad (mAP 60.4%, Rank-1 88.9%). The training log confirms that SGD struggled to converge within 10 epochs — training accuracy reached only around 23%, with Xent remaining above 3.9 throughout. This indicates that the standard SGD step size, even with the tuned LR of 0.0001, was insufficient for effective learning within this short training schedule.

AMSGrad, a variant of Adam with a guaranteed non-increasing effective learning rate, is better suited to this task as it adapts per-parameter learning rates and handles sparse gradients more robustly. SGD would likely require a longer training schedule and careful momentum/LR decay tuning to reach comparable performance. For this 10-epoch constraint, AMSGrad is clearly the superior optimiser.

Optimizer	mAP	Rank-1	Rank-5	Rank-10
<u>AMSGrad (best from 3.2)</u>	<u>60.4%</u>	<u>88.9%</u>	<u>95.3%</u>	<u>97.3%</u>
SGD	28.8%	55.5%	72.1%	79.0%

SECTION 4: SUMMARY – HYPERPARAMETER TUNING

Section	Hyperparameter	Best Configuration	mAP	Rank-1
1	Architecture	ResNet-50	52.4%	83.8%
2	Data Augmentation	Default + Colour Jitter + Random Erasing	52.5%	83.6%
3.1	Learning Rate	0.0001	59.3%	86.2%
3.2	Batch Size	32	60.4%	88.9%
3.3	Optimizer	AMSGrad	60.4%	88.9%

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